CIE Review of the Alaska-wide Sablefish Stock Assessment

Performance Work Statement (PWS)
National Oceanic and Atmospheric Administration (NOAA)
NOAA Fisheries
Center for Independent Experts (CIE) Program
External Independent Peer Review

June 16 - 18, 2026

Background

NOAA Fisheries is mandated by the Magnuson-Stevens Fishery Conservation and Management Act, Endangered Species Act, and Marine Mammal Protection Act to conserve, protect, and manage our nation's marine living resources based upon the best scientific information available (BSIA). NOAA Fisheries science products, including scientific advice, are often controversial and may require timely scientific peer reviews that are strictly independent of all outside influences. A formal external process for independent expert reviews of the agency's scientific products and programs ensures their credibility. Therefore, external scientific peer reviews have been and continue to be essential to strengthening scientific quality assurance for fishery conservation and management actions.

Scientific peer review is defined as the organized review process where one or more qualified experts review scientific information to ensure quality and credibility. These expert(s) must conduct their peer review impartially, objectively, and without conflicts of interest. Each reviewer must also be independent from the development of the science, without influence from any position that the agency or constituent groups may have. Furthermore, the Office of Management and Budget (OMB), authorized by the Information Quality Act, requires all federal agencies to conduct peer reviews of highly influential and controversial science before dissemination, and that peer reviewers must be deemed qualified based on the OMB Peer Review Bulletin standards[1].

Sablefish (*Anoplopoma fimbria*) are a long-lived (90+ years), highly mobile, sexually dimorphic demersal species that live at depths greater than 200m (as adults). In Alaska Federal waters, sablefish are assessed and managed as a single population with the total quota apportioned across 6 management regions (Aleutian Islands, Bering Sea, Western Gulf of Alaska, Central Gulf of Alaska, Western Yakutat, and Southeast Outside) and among two fishing sectors (i.e., fixed gear and trawl gear). Since rationalization (i.e., implementation of individual fishing quotas, IFQs) occurred in the 1990s, sablefish have been one of the most valuable finfish resources in Alaska ($124 million in ex-vessel value in 2022). Catch advice for sablefish is based on a statistical catch-at-age integrated stock assessment model fit to fishery-independent survey data and fishery-dependent catch-age data. The sablefish assessment is reviewed annually by the North Pacific Fishery Management Council's (NPFMC) Science and Statistical Committee (SSC) as part of the quota specification process, and last underwent a CIE review in 2016.

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[1] https://www.whitehouse.gov/wp-content/uploads/legacy_drupal_files/omb/memoranda/2005/m05-03.pdf

Scope

Since the last CIE review (i.e., in 2016), major changes have occurred in the sablefish resource, fishery, longline survey, and assessment model. Specifically, a series of extreme, above average recruitment events has led to rapid rebuilding of the resource and increasing quotas. Resultant landings of predominantly small, lower value fish have led to market saturation, strong reductions in landed value, and exploration of management options that loosen full retention requirements. Concomitantly, the primary directed fixed gear fleet has undergone rapid gear changes, where a predominantly hook-and-line fishery has transitioned to a longline pot fishery since 2017 (i.e., >85% of sablefish catch in the fixed gear fishery is now taken by pot gear), which has helped to reduce whale depredation.

Additionally, the fishery-independent, cooperative research-based NOAA longline survey, which targets sablefish and serves as the primary index of abundance for the assessment, underwent major survey design changes starting in 2025. Because the survey has traditionally used a cost-recovery approach, whereby survey catch is sold by the fishing company that owns the vessel performing the survey, the decline in sablefish markets prevented the survey from being undertaken in 2024 for the first time in its 30+ year history. Historically, the fixed station survey has sampled the entire Gulf of Alaska (GOA) annually, as well as either the Bering Sea (BS, odd years) or Aleutian Islands (AI, even years). Starting in 2025 a new survey design was enacted to reduce overhead and ensure a fishing company would accept the survey contract, wherein the Gulf of Alaska is now surveyed in odd years (starting in 2025), and the BS and AI are surveyed in even years (starting in 2026). Although the new survey design ensures the ability to continue the longline survey, it results in only ~50% of total stock abundance being surveyed in a given year. Moreover, it raises important challenges for the panmictic assessment in terms of how to integrate survey data that only partially samples the population being assessed.

There have also been major changes to the assessment model since the last CIE review. Most importantly, a new R-TMB based package, the SPoRC (Stochastic Population Over Regional Components) assessment model, has been adopted, which replaces the previous Automatic Differentiation Model Builder (ADMB) based assessment model. Additionally, a number of research-oriented assessment models have been developed exploring alternative spatial structures (e.g., spatially-explicit 3 and 5 region models that estimate movement among regions, as well as a spatially-implicit single region fleets-as-areas, FAA, model), all of which demonstrate promise for dealing with spatial complexities and data gaps that are not well resolved in the single region panmictic assessment. In 2026, the assessment team transitioned from the single region panmictic assessment to the FAA model, because the FAA model could better handle the new longline survey design (i.e., fit the data by region as separate fleets) and implicitly account for ontogenetic movement dynamics (i.e., through selectivity and availability assumptions). Thus, for the purpose of the CIE review, feedback on the FAA model is being requested, particularly with regard to appropriate model structure and parametrization. To help make determinations on the BSIA to assess the Alaska sablefish population, reviewers will be provided the results of closed loop simulation analyses and assessment model runs applied to the sablefish data (including model outputs and diagnostics).

The overarching goals of this review process are to:

1. Ensure the FAA stock assessment supports the development of robust advice for use by the North Pacific Fishery Management Council based on the BSIA;
2. Provide an independent external review of this stock assessment to confirm that it meets the mandates of the Magnuson-Stevens Fisheries Conservation and Management Act (MSA) and other legal requirements;
3. Understand and identify appropriate FAA model structure and fits to key data sources (indices, landings, and age compositions);
4. Identify research needed to improve the assessment and advice for fishery managers.

The specified format and contents of the individual peer review report are found in **Annex 1**. The Terms of Reference (ToRs) of the peer review are included in **Annex 2**. The tentative agenda of this in-person panel review meeting is included in **Annex 3**.

Requirements

NOAA Fisheries requires 3 reviewers to conduct an impartial and independent peer review in accordance with this Performance Work Statement (PWS), OMB Guidelines, and the ToRs below. Modifications to this Performance Work Statement (PWS) and ToRs cannot be made during the peer review, and the Contracting Officer's Representative (COR) and the CIE contractor shall approve any modifications prior to the peer review. All ToRs must be addressed in each reviewer's report.

The CIE reviewers should have expertise in statistical sex- and age-structured integrated stock assessment models. In particular, experience developing spatially explicit and spatially implicit (i.e., fleets-as-areas) assessment models is imperative. Moreover, understanding of stock assessment good practices are required. Working knowledge of the RTMB programming language, a basic understanding of closed loop simulation, and a background in survey design and/or the use of fishery-independent survey data in stock assessments would all be beneficial.

The chair for this meeting, who is in addition to the CIE reviewer, will not be provided by the CIE. Although the chair will be participating in this review, the chair's participation (e.g., labor and travel) is not covered by this contract.

Tasks for Reviewers

The CIE reviewer shall complete the following tasks in accordance with the PWS and Schedule of Milestones and Deliverables herein.

Pre-review Background Documents: A minimum of two weeks before the peer review, the NOAA Fisheries Project Contact will send (by electronic mail or make available at an FTP site/GitHub Repository) to the CIE reviewers the necessary background information and reports for the peer review. In the case where the documents need to be mailed, the NOAA Fisheries Project Contact will consult with the CIE on where to send documents. The CIE reviewers are responsible only for the pre-review documents that are delivered to the reviewer in accordance with the PWS scheduled deadlines specified herein.

The document list is separated into 'summary/background' and 'for review' documents. The CIE reviewers are *encouraged to peruse* the 'summary/background' documents for information on data inputs, previous assessment structures and associated reviews, and ongoing research. However, the CIE reviewers *must review* all of the 'for review' documents in preparation for the peer review, as these contain the results of the primary analyses and model runs that will be discussed during the review panel.

Available documents (available from the public sablefish CIE GitHub repository, as they become available) include:

For Review (available here)

- **2026 Assessment of the Sablefish Stock in Alaska: 2026 Center for Independent Experts (CIE) Review Document (here)**
- **Appendix 3A. The Spatial Population over Regional Components (SPoRC) Model: Technical Description of the Assessment Model for Sablefish (here)**
- **Appendix 3B. Model Building and Bridging Exercise for the Sablefish Single Region Models (Bridging from Model *25.12* to *26.1_1Reg*) (here)**
- **Appendix 3C. Development of a Fleets-as-Areas (FAA) Stock Assessment Model for Alaska Sablefish (*Model 26.2_FAA*) (here)**
- **Appendix 3D. Results of Spatial Stock Assessment Models for Alaska Sablefish (here)**
- **Appendix 3E. Evaluating Alternative Assessment Structures for Alaska Sablefish Under Survey Design Changes via Closed-Loop Simulation (here)**

Please note that outputs of all analyses used as part of the CIE review, including input data, simulation outputs, final proposed assessment models, and all sensitivity assessment models, will be made available on the sablefish CIE GitHub repository

Panel Review Meeting: The CIE reviewers shall conduct the independent peer review in accordance with the PWS and ToRs and shall not serve in any other role unless specified herein. The CIE reviewers shall actively participate in a professional and respectful manner as members of the meeting review panel, and their peer review tasks shall be focused on the ToRs as specified herein. The NOAA Fisheries Project Contact is responsible for any meeting arrangements (e.g., conference room reservations, video conferencing logistics, etc.).

Contract Deliverables - Independent CIE Peer Review Report: Each CIE reviewer shall complete an independent peer review report in accordance with this PWS. Reviewers are not required to reach a consensus, and a summary report *will not* be produced. Each CIE reviewer shall complete their independent peer review according to the required format and content as described in **Annex 1**. Each CIE reviewer shall complete the independent peer review addressing each ToR as described in **Annex 2**. The tentative agenda of the panel review meeting is in **Annex 3**. Each CIE reviewer will deliver their reports according to the specified milestone dates.

Foreign National Security Clearance

When reviewers participate during a panel review meeting at a government facility, the NOAA Fisheries Project Contact is responsible for obtaining the Foreign National Security Clearance approval for reviewers who are non-US citizens. For this reason, the reviewers shall provide requested information (e.g., first and last name, contact information, gender, birth date, passport number, country of passport, travel dates, country of citizenship, country of current residence, and home country) to the Project Contact for the purpose of their security clearance, and this information shall be submitted at least two weeks in advance. For additional information, please see the following link: https://www.commerce.gov/osy/programs/foreign-access-management. All CIE reviewers must have a REAL ID-compliant form of identification to access Federally owned or leased facilities. The contractor is required to use all appropriate methods to safeguard Personally Identifiable Information (PII).

Place of Performance
The place of performance shall be Juneau, AK.

Period of Performance
The period of performance shall be from the time of award through July 2026. The CIE reviewers' duties shall not exceed **14** days to complete all required tasks.

Schedule of Milestones and Deliverables
The contractor shall complete the tasks and deliverables in accordance with the following schedule:

Within two weeks of award	Contractor secures participation of the CIE reviewers.
At least two weeks prior to the panel review meeting	Pre-review documents are provided to the reviewers.
June 16 - 18, 2026	The reviewers participate and conduct an independent peer review during the panel review meeting.
Approximately 2 weeks later	Contractor receives draft reports.
Within 3 weeks of receiving draft reports	Contractor submits final reports to the US Government.

Applicable Performance Standards
The acceptance of the contract deliverables shall be based on three performance standards: (1) The reports shall be completed in accordance with the required formatting and content in **Annex 1**; (2) The reports shall address each ToR as specified **Annex 2**; and (3) The reports shall be delivered as specified in the schedule of milestones and deliverables.

Travel

All travel expenses shall be reimbursable in accordance with Federal Travel Regulations (Travel resources | GSA), and all contractor travel must be approved by the COR prior to the actual travel. Any travel conducted prior to the receipt of proper written authorization from the COR will be done at the Contractor's own risk and expense. International travel is authorized for this contract. Travel is not to exceed $13,000.00.

Confidentiality and Data Privacy
This contract may require that services contractors have access to Privacy Information. Services contractors are responsible for maintaining the confidentiality of all subjects and materials and may be required to sign and adhere to a Non-disclosure Agreement (NDA).

NOAA Fisheries Project Contact:
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Annex 1: Peer Review Report Requirements

1. The CIE independent report shall be prefaced with an Executive Summary providing a concise summary of the findings and recommendations as specified in the ToR and specify whether the science reviewed is the best scientific information available.

2. The main body of the reviewer report shall consist of the following sections: Background; Description of the Individual Reviewer's Role in the Review Activities; Summary of Findings for each ToR in which the weaknesses and strengths are described; and Conclusions and Recommendations in accordance with the ToRs.

   a. Each reviewer should describe in their own words the review activities completed during the panel review meeting, including providing a brief summary of findings as well as the science, conclusions, and recommendations.

   b. Each reviewer should discuss their independent views on each ToR even if these were consistent with those of other panelists, and especially where there were divergent views.

   c. Each reviewer should elaborate on any points that they feel might require further clarification.

   d. Each independent CIE report shall be a stand-alone document through which others can understand the weaknesses and strengths of the science reviewed and shall be an independent peer review of each ToRs.

3. The reviewer report shall include the following appendices:

   Appendix 1: Bibliography of materials provided for review.
   Appendix 2: A copy of the CIE Performance Work Statement.
   Appendix 3: Panel Membership or other pertinent information from the panel review meeting.

Annex 2: Terms of Reference (ToRs) for the Peer Review

Review of the Alaska-wide Sablefish Assessment

CIE reviewers are contracted to complete their independent peer review based on the ToRs. Therefore, the CIE-NOAA Fisheries review and approval process is based on whether the CIE independent reports addressed each of the ToRs.

1) Review the disseminated documents and analyses, then provide feedback on whether the fleets-as-areas (FAA) assessment provides a suitable basis for operational management advice, particularly in comparison to the current single region panmictic assessment.
   a) Provide recommendations on the most appropriate parametrization of fleet structure (number of fleets per fishery/survey, as a proxy for regions) in the FAA model.

2) Highlight potential strengths and weaknesses of the FAA assessment, including model assumptions, estimates, fits to data (survey indices and age compositions along with fishery catch-at-age data), model diagnostics, and major sources of uncertainty.
   a) Indicate whether the model provides the best scientific information available for estimating current stock status, projecting future abundance, and determining Overfishing Levels (OFLs) and Acceptable Biological Catches (ABCs).

3) Provide research recommendations for improving the existing research-oriented sablefish spatial (i.e., 3 and 5 region) assessment models, especially in the context of potential future usage in a management context.

Reviewers must provide details for the basis of their responses to the review ToRs.

Annex 3: *Tentative* Agenda for the CIE Review of the Alaska-wide Sablefish Assessment

North Pacific Fishery Management Council
NOAA Alaska Fisheries Science Center, Auke Bay Laboratories
17109 Pt. Lena Loop Road
Juneau, AK 99801

June 16-18, 2026

Contact: Chris Lunsford (Chris.Lunsford@noaa.gov)

Schedule

(this agenda is *DRAFT* and subject to change)

All times below are Alaska Daylight Time
Daily breaks at ~10:30 and ~15:30, Lunch 12:30-14:00
Tuesday-Thursday, 09:00 – 17:00 AKDT

Tuesday, June 16, 2026

A. **Welcome and Introductions (30min)**
B. **Sablefish Background (Management, Biology, and Data Inputs; 3hrs)**
C. **General Assessment Model Structure (ToR 1; 3hrs)**

Wednesday, June 17, 2026

A. **Reconvene and Review (30min)**
B. **Summary of Closed Loop Simulations (3hrs)**
C. **Comparing Assessment Performance and Diagnostics (ToRs 1/2; 3hrs)**

Thursday, June 18, 2026

A. **Reconvene and Review (30min)**
B. **Comparing Assessment Performance and Diagnostics *Cont'd* (ToR 1/2; 3hrs)**
C. **Spatial Model Configurations and Diagnostics (ToR 3; 3hrs)**

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